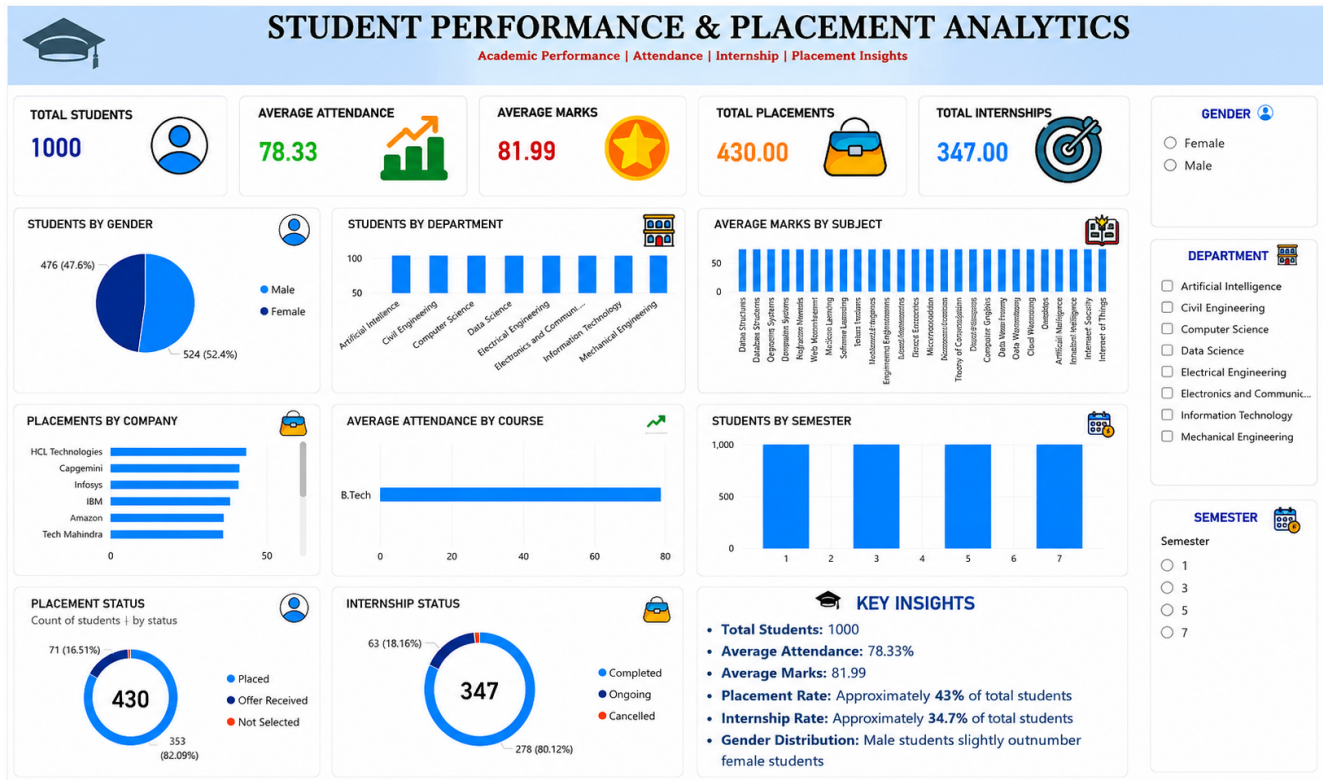


STUDENT PERFORMANCE & PLACEMENT ANALYTICS

Power BI Data Analytics Project Documentation



Project Focus: Student Academic Performance • Attendance • Placement • Internship Analysis

Professional portfolio documentation

1. Project Overview

The Student Performance & Placement Analytics project is an interactive business intelligence dashboard designed to analyze student academic performance and career-related outcomes. The project combines data related to students, departments, subjects, marks, attendance, internships, placements, and assignments.

The main objective is to transform raw relational data into meaningful insights using data modeling, KPI cards, interactive charts, slicers, and a professionally designed Power BI dashboard.

2. Project Objectives

- Analyze the overall student population and academic performance.
- Measure average attendance and identify engagement patterns.
- Compare student distribution across departments, subjects, and semesters.
- Analyze placement outcomes and company-level placement activity.
- Track internship participation and internship status.
- Create interactive filtering using gender, department, and semester slicers.

3. Tools & Technologies

Tool / Technology	Purpose
PostgreSQL	Relational data management and table handling
Power BI	Data modeling, visualization, and dashboard development
CSV Files	Dataset import/export and structured data management
Data Modeling	Relationships between multiple related tables
KPI Development	Aggregations for important performance indicators

4. Dataset and Data Model

The project uses a multi-table relational structure. Student information is connected with academic, attendance, internship, placement, department, subject, and assignment records. Relevant relationships were created using key columns such as student identifiers and department identifiers.

- students — student profile and demographic information
- departments — department-level information
- subjects — subject and department mapping
- marks — student academic marks and grades
- attendance — attendance percentage and classes attended
- internships — internship participation and status
- placements — company, job role, package, date, and placement status
- assignments — assignment-related information

5. Dashboard Features

- KPI cards for Total Students, Average Attendance, Average Marks, Total Placements, and Total Internships
- Students by Gender analysis
- Students by Department comparison
- Average Marks by Subject visualization
- Placements by Company analysis
- Average Attendance by Course comparison
- Students by Semester visualization
- Placement Status and Internship Status charts
- Interactive slicers for Gender, Department, and Semester
- Key Insights section for executive-level summary

6. Key Performance Indicators

KPI	Result	Description
Total Students	1,000	Overall student population analyzed
Average Attendance	78.33%	Average attendance level
Average Marks	81.99	Overall academic performance
Total Placements	430	Placement outcomes shown in the dashboard
Total Internships	347	Internship outcomes shown in the dashboard

7. Key Insights

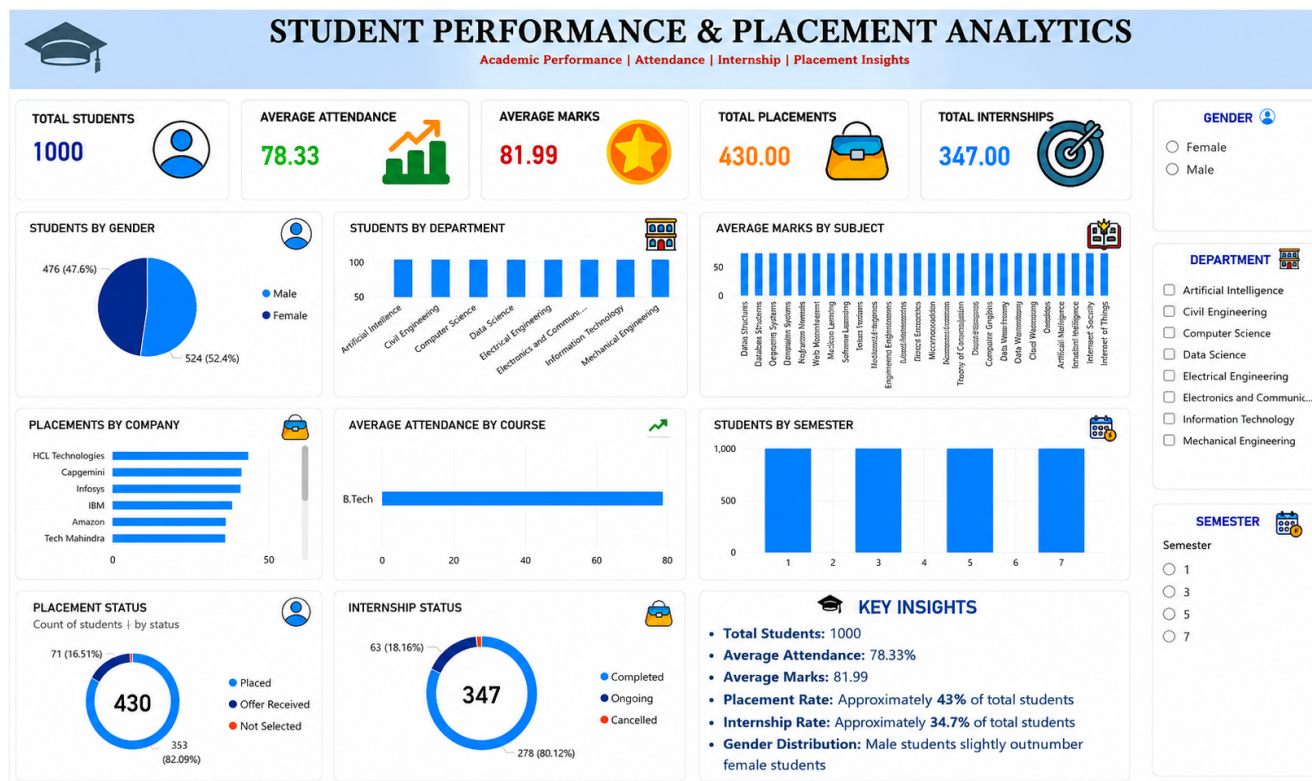
- The dashboard analyzes a total population of 1,000 students.
- Average attendance is 78.33%, providing a clear view of overall student engagement.
- Average marks are 81.99, indicating strong academic performance in the analyzed data.
- Placement outcomes total 430, representing approximately 43% when compared with the total student count used in the dashboard summary.
- Internship outcomes total 347, representing approximately 34.7% when compared with the total student count used in the dashboard summary.
- Gender distribution is relatively balanced, with male students slightly outnumbering female students in the displayed data.
- Interactive filters allow users to explore the data by gender, department, and semester.

8. Project Workflow

Step	Stage	Activity
1	Data Preparation	Organized and prepared student-related datasets.
2	Database Handling	Managed relational tables using PostgreSQL.
3	Data Import	Loaded relevant datasets into Power BI.
4	Data Modeling	Created and verified relationships between related tables.
5	KPI Development	Calculated student, attendance, marks, placement, and internship metrics.
6	Visualization	Built charts, cards, slicers, and insights.
7	Formatting	Applied consistent colors, spacing, layout, and design.

9. Final Dashboard Preview

The final dashboard provides a consolidated view of student academic performance, attendance, placements, internships, and demographic patterns.



10. Conclusion

This project demonstrates an end-to-end data analytics workflow, from relational data preparation and modeling to KPI development, visualization, and insight generation. The final dashboard enables users to explore important student performance and career-related metrics through an interactive and professionally structured interface.

Skills Demonstrated: Data Analytics, Power BI, PostgreSQL, Data Modeling, Data Visualization, KPI Development, Dashboard Design, and Business Insight Generation.